

Repo Layout & Decoupling

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Master technical specification — Volume 6 (Deployment, Tooling & Operations), nano-detail document. Scope: the **physical repository layout** of HelixVPN — the Phase-0/1 working monorepo `helixvpn/`, the planned `vasic-digital/*` decoupled reusable-component repos, the extraction sequencing, the decoupling invariants (§11.4.28), snake_case naming (§11.4.29), catalogue-first reuse (§11.4.74), the dependency-from-root layout (§11.4.28C), and the overview-level wiring points for the already-incorporated submodules/ members (`helix_design/containers/helix_qa/challenges/docs_chain`). This deepens [05-overview §1/§2/§3/§6] into an implementation-ready repo spec. It is a SPEC: trees, manifests, and procedures are illustrative of the contract, not the shipping code (2–3 refinement passes follow). Detailed ecosystem wiring lives in `helix-ecosystem-integration.md`; this doc gives the seam, not the depth.

Evidence cited inline by id: [05_OV §N] = 05-repo-layout-tooling-and-helix-ecosystem.md; [04_ARCH §N] = 04_VPN_CLD/HelixVPN-Architecture-Refined.md §11 (repo layout); [04_P1] = HelixVPN-Phase1-MVP.md §2 (repo skeleton); [04_P0] = HelixVPN-Phase0-Spike.md (surviving Transport trait + FFI seams); [SYNTHESIS §N] = v09-research/_SYNTHESIS.md §6 (reuse pillars). Unproven facts are marked **UNVERIFIED** per §11.4.6 — never fabricated.

0. What this document owns (and does not)

This document owns the **repository topology decisions**: the monorepo-now / extract-later strategy, the file-by-file monorepo tree, the six decoupled-repo public-contract definitions, the extraction order +

procedure, the per-repo scaffold checklist, the `helix-deps.yaml` dependency manifests, the release-prefix application to every repo, and the overview-level map of which submodules/ member binds where.

It does **not** own: the codegen pipeline internals (that is `codegen-pipeline.md`); the `helixvpctl` command spec (that is `helixvpctl.md`); the deploy substrates (`podman-quadlets.md`, `docker-compose.md`, `kubernetes.md`); the depth of each submodule's integration (`helix-ecosystem-integration.md`); the design-system repo's internals (Volume 10, `vasic-digital/helix_design`). Those are referenced, never redefined.

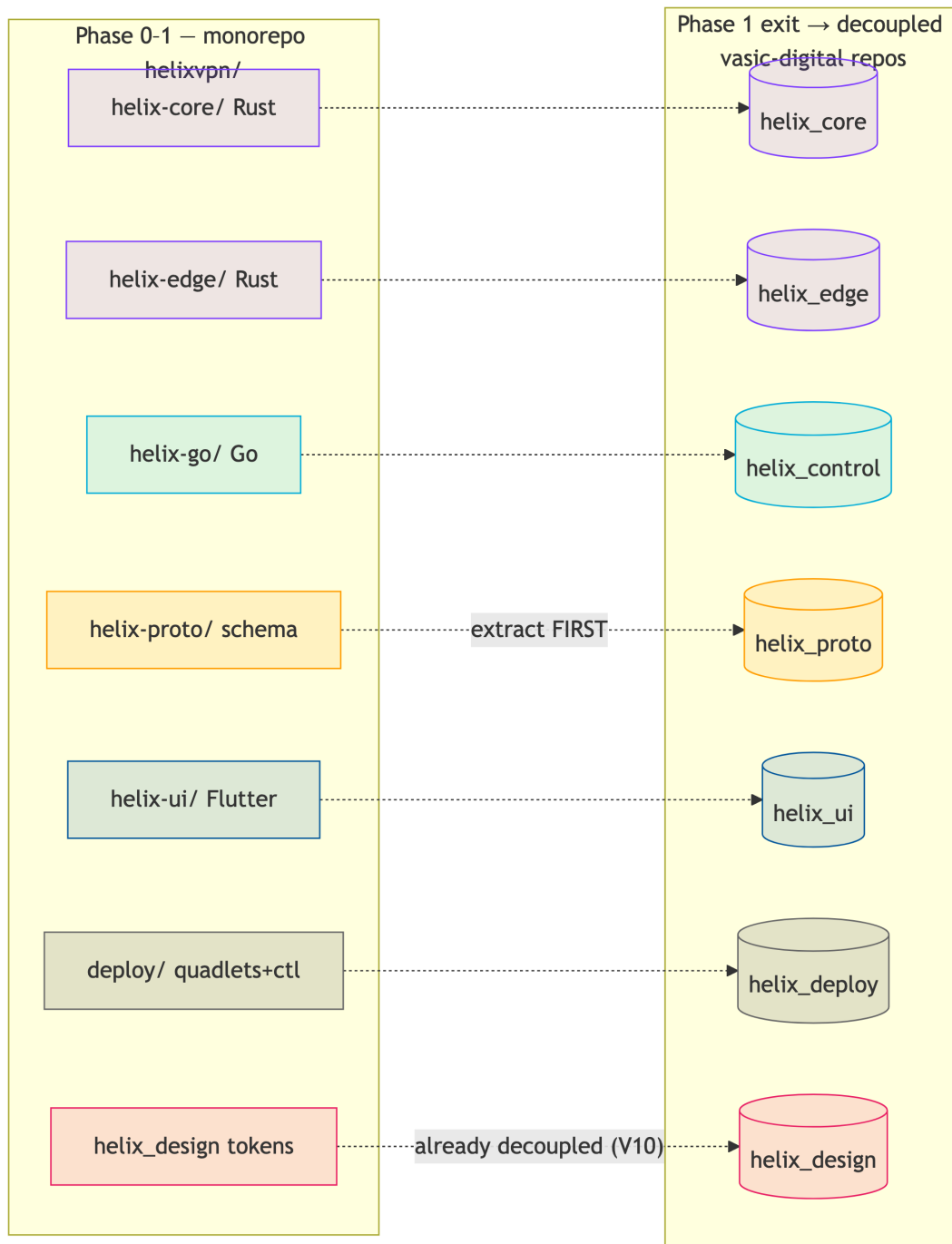
1. The monorepo-now / extract-later strategy

1.1 The tension and its resolution

The constitution wants **decoupled, reusable, project-not-aware components**, each in its own repo (§11.4.28/.74). The MVP wants **velocity and atomic cross-cutting changes** — one PR that touches `.proto` + Go + Rust + Dart in lock-step. These pull in opposite directions only when treated as simultaneous obligations; they are not. The binding resolution [05_OV §1.1]:

Phase 0 → Phase 1: one working monorepo `helixvpn`/ whose internal workspace boundaries are *already* the future repo boundaries.
Phase-1 exit gate: extract the stabilized pillars into standalone `vasic-digital` repos and re-consume them as flat submodules. Nothing is extracted before its public surface stops churning (§11.4.124 — don't split before the seams are known).

This applies, at the *repository* level, the same discipline the control plane applies at the *package* level — "one Go binary, many packages; package boundaries == future service boundaries" [04_ARCH §4.1]. It honors §11.4.28 (the boundaries exist from day one, merely co-located) without paying the multi-repo coordination tax during the highest-churn phase.



helix_design is the exception (Volume 10, operator mandate 2026-06-25, §11.4.162): the design system is a decoupled vasic-digital/helix_design submodule **from day one**, not an extract-later pillar, because it is mandated reusable by any future app and its token contract is independent of HelixVPN code churn. The other six follow the extract-later path.

1.2 Extraction order (most-stable-first)

Order	Monorepo dir	Extracted repo (vasic-digital)	Trigger to extract
1	helix-proto/	helix_proto	First frozen WatchNetworkMap version [04_P1 §3] — the schema is the contract; it must stop drifting before generated clients can stabilize.
2	helix-core/	helix_core	iOS memory gate G3 passes and the helix-ffi C-ABI/UniFFI surface freezes [04_P0].
3	helix-edge/	helix_edge	Immediately after helix_core (it path-depends on helix-transport).
4	helix-ui/	helix_ui	Riverpod-over-status-stream contract + runHelixApp flavors stable by MVP DoD [04_UI].
5	helix-go/	helix_control	Largest churn surface (policy compiler, coordinator) — extract last among code.
6	deploy/	helix_deploy	Image names + the env contract freeze (so the quadlet/compose/k8s generators reference stable names).

Each extraction is a **visible commit** citing §11.4.35 ("Lifted from helixvpn monorepo to vasic-digital/helix_proto per §11.4.28"), preserves history via `git filter-repo --path helix-proto/`, and immediately re-consumes the new repo as a flat submodule (§4) [05_OV §1.2].

2. Monorepo layout (helixvpn/) — file-by-file

Directory and file names are **lowercase snake_case** where they are ours to name (§11.4.29); language-mandated layouts (Rust `crates/`, Dart `lib/`, Cargo/Go file conventions) follow their ecosystem (the §11.4.29 technology-exception) [05_OV §2].

helixvpn/	# umbrella working repo
(this repository's product code)	
├ Cargo.toml	# Rust workspace root –
members: helix-core/crates/*, helix-edge	
├ go.work	# Go workspace –

```

uses ./helix-go (+ extracted submodules later)
├─ melos.yaml # Dart/Flutter monorepo
manager (helix-ui packages)
├─ buf.yaml buf.gen.yaml buf.lock # schema-first codegen
config (codegen-pipeline.md)
├─ Makefile # local task runner
(gen/build/test/deploy) – NO CI (§11.4.156)
├─ .gitignore # build
artifacts, .env, *.db-wal/-shm (§11.4.30)
├─ .env.example # HELIX_RELEASE_PREFIX,
infra creds template (§11.4.77/.151)
├─ helix-deps.yaml # submodule dependency
manifest (§11.4.31)
|
├─ helix-proto/ # — reuse pillar 1:
the contract —
|   └─ proto/helix/v1/ # coordinator.proto,
enrollment.proto, types.proto
|   └─ openapi/helix-rest.v1.yaml # Console/Access/
Connector REST surface
|   └─ gen/ # GENERATED
(gitignored) → go/ dart/ rust/ ts/ (codegen-pipeline.md)
|
├─ helix-core/ # — reuse pillar 2a:
Rust client+connector core —
|   └─ Cargo.toml
|   └─ crates/ # helix-transport,
helix-wg, helix-reconcile, # helix-netshield,
helix-ffi
|
├─ helix-edge/ # — reuse pillar 2b:
Rust gateway data-plane edge —
|   └─ Cargo.toml # path-dep on helix-
core/crates/helix-transport (D5)
|   └─ src/ # quinn+h3 MASQUE
termination + kernel-WG fast path
|
├─ helix-go/ # — the Go control
plane (modular monolith) —
|   └─ go.mod # module helixvpn/
helix-go # control-plane
|   └─ cmd/helixd/ # operator CLI
binary (all packages wired) # identity registry
|   └─ cmd/helixvpnctl/ #
(Cobra) – helixvpnctl.md #
|   └─ internal/ #
ipam pki policy coordinator events telemetry api store
|
├─ helix-ui/ # — reuse pillar 3:
Flutter —
|   └─ packages/ #
helix_design(consumes V10), helix_core_ffi, helix_api_client,
| # app_access,

```

```

app_connector, app_console
├─ shims/                                # per-platform
TunnelPlatform providers (ONLY platform code)
├─ apple/ android/ windows/ linux/      # Phase 0–2
├─ harmonyos/ aurora/                  # Phase 3
├─ deploy/                              # — deployment
substrate (→ helix_deploy) —
├─ quadlets/                            # *.pod *.container
*.network *.volume (CANONICAL, §11.4.76)
├─ compose/docker-compose.yaml          # Docker-operator
equivalence (docker-compose.md)
├─ k8s/                                  # plain manifests +
kustomize (kubernetes.md)
├─ terraform/                           # gateway VPS
provisioning
├─ grafana/                             # dashboards-as-code
(counters only)
├─ submodules/                          # already-incorporated
Helix ecosystem (§5)
├─ helix_design/ containers/ docs_chain/ helix_qa/
challenges/ security/
├─ vision_engine/ panoptic/ doc_processor/
├─ llm_orchestrator/ llm_provider/ llms_verifier/ (dev-loop
only – §5.2)
├─ constitution/                        # HelixConstitution
submodule (canonical root, §11.4.35)

```

2.1 Workspace manifests (the boundary-as-config seams)

Three workspace managers keep the internal boundaries crisp so extraction is a git filter-repo

- re-consume, never a refactor [05_OV §2.1]:
- **Cargo.toml** (Rust) — one resolver, one shared lockfile, **path deps** so helix-edge reuses the helix-transport crate byte-for-byte (the D5 "shares helix-transport" property [SYNTHESIS §3]). The workspace members ARE the future helix_core crate set + helix_edge.
- **go.work** (Go) — keeps helixd + helixvpctl in one module pre-extraction; post- extraction the containers/challenges submodules join via replace directives during dev + pinned SHAs in release (§11.4.76 consumption rule).
- **melos.yaml** (Dart/Flutter) — governs the helix-ui package tree; runHelixApp(flavor,...) produces all three apps from packages/app_* [04_UI]. The Console is the only Web build and omits helix_core_ffi.

A Cargo member, a Go package, or a Melos package is a future-repo boundary expressed today as a workspace entry — this is the mechanical form of "the seams exist from day one".

3. The decoupled reusable-component repos (the §11.4.28/.29/.74 target)

3.1 The seven repos and their public contracts

Six are created **after** the spec stabilizes (per §1.2 order); `helix_design` exists from day one (Volume 10). Each is its own GitHub **and** GitLab repo (§2.1 multi-mirror), `snake_case` (§11.4.29), **flat** under the consumer (no nested own-org `.gitmodules` chains — §11.4.28C), with an `upstreams/recipe` dir + `install_upstreams` run on clone (§11.4.36) [05_OV §3.1].

Repo	Language	Public surface (the ONLY thing consumers depend on)	Consumed by
<code>helix_proto</code>	proto + OpenAPI	<code>proto/helix/v1/*.proto</code> , <code>openapi/helix-rest.v1.yaml</code> ; ships generated Go/Dart/Rust/TS as release artifacts	core, edge, control, ui
<code>helix_core</code>	Rust	Transport trait, <code>helix-wg</code> API, <code>helix-ffi</code> C-ABI/UniFFI surface	edge, Access/Connector apps, shims
<code>helix_edge</code>	Rust	gateway data-plane binary + <code>EdgeConfig</code> (no lib API beyond config)	deploy
<code>helix_control</code>	Go	<code>helixd</code> binary + REST/proto contracts (the internal packages stay private)	deploy, <code>helixvpncctl</code>
<code>helix_ui</code>	Dart/Flutter	<code>helix_design</code> consumption + <code>runHelixApp()</code> entry + Riverpod status-stream contract	the 3 apps

Repo	Language	Public surface (the ONLY thing consumers depend on)	Consumed by
helix_deploy	Go+YAML+HCL	helixvpnctl binary + quadlet/compose/k8s generators	operators
helix_design	tokens + per-platform theme packages	OpenDesign-emitted token bundles (JSON/CSS/Dart/Swift/Kotlin/ArkTS/C-Qt) + component specs (Volume 10, §11.4.162)	helix_ui + any future app

3.2 Decoupling invariant (§11.4.28B) — the load-bearing rule

None of these may hardcode HelixVPN-specific hostnames, asset names, tenant assumptions, or brand strings. `helix_core` is a generic WG-over-pluggable-transport engine; `helix_proto` is a generic overlay-coordination schema; `helix_design` is OpenDesign tokens with brand values injected, not baked. Project specifics enter ONLY via **config injection** — env var, config struct, constructor parameter — never a hardcoded reach back into HelixVPN [05_OV §3.1].

The gate CM-OWNED-SUBMODULE-DECOUPLING (§11.4.28) greps each submodule diff for parent-project context; a hostname/asset/tenant literal inside a submodule fails it. **UNVERIFIED:** the exact deny-list of "parent context" tokens is set per-repo at extraction time, not fixed here.

3.3 Per-repo scaffold checklist (applied to all seven)

```
<repo>/
├─ README.md  CLAUDE.md  AGENTS.md  GEMINI.md  QWEN.md  #
inherit constitution (§11.4.35/.157)
├─ constitution/  #
HelixConstitution submodule (canonical root)
├─ upstreams/{github.sh,gitlab.sh}  #
remote recipes for install_upstreams (§11.4.36)
├─ helix-deps.yaml  # own-
org Git SSH deps (§11.4.31)
├─ .gitignore  .env.example  #
§11.4.30/.77
├─ docs/  #
Status.md + Status_Summary.md + guides (§11.4.45/.56)
├─ └─ .docs_chain/contexts/<repo>.yaml  #
```



```
docs_chain export + DB sync (§11.4.106)
└─ (language tree)
```

3.4 helix-deps.yaml — the dependency manifest (§11.4.31)

helix_edge is the only code repo with an own-org build-time dependency, so its manifest is the worked example [05_OV §3.2]:

```
# helix_edge/helix-deps.yaml
schema_version: 1
deps:
  - name: helix_core
    ssh_url: git@github.com:vasic-digital/helix_core.git
    ref: main
    why: "helix-edge reuses the helix-transport crate byte-for-
         byte (D5)"
    layout: flat
transitive_handling:
  recursive: true
  conflict_resolution: operator-required
language_specific_subtree: false
```

helix_ui's manifest declares helix_design the same way; helix_control's declares containers + challenges. The manifest is what lets a fresh consumer reconstruct the dependency graph at its own root (§11.4.31) without nested own-org chains.

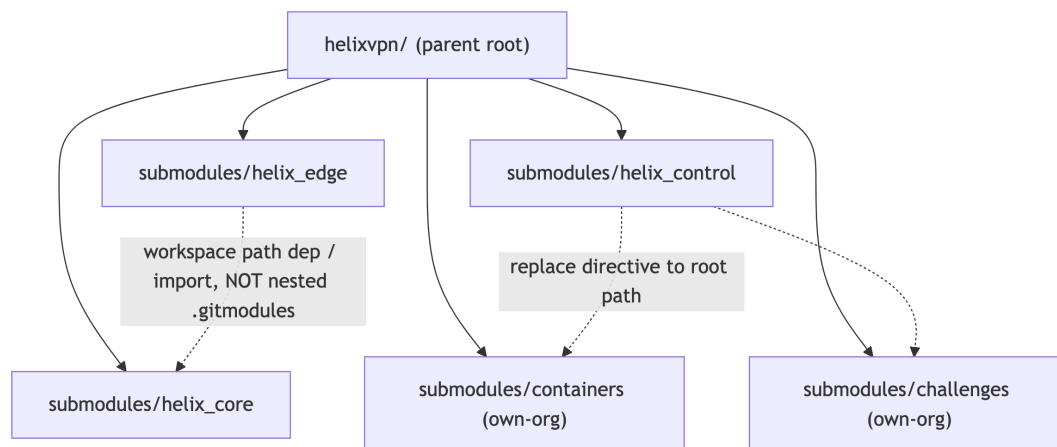
4. Dependency-from-root layout (§11.4.28C) — no nested own-org chains

Every dependency consumed by an owned submodule MUST be reachable from the **parent project's root**, NOT nested inside the submodule [05_OV §3, §11.4.28C]:

```
helixvpn/                                # parent root
├─ submodules/helix_core/                # dependency at the root
├─ submodules/helix_edge/                # consumer — reaches helix_core via
go.work/Cargo path, NOT a nested submodule
└─ submodules/helix_control/             # consumer — reaches containers/
challenges via the root, not nested
```

Forbidden: helix_edge/.gitmodules pulling in helix_core as a nested own-org submodule. **Required:** helix_core sits at helixvpn/submodules/helix_core, and helix_edge reaches it via the workspace path dep (Cargo) / the documented import resolver — a flat graph the CM-OWNED-SUBMODULE-

LAYOUT gate audits (§11.4.28). Third-party submodules (the §11.4.74 no-match → vendor path) are exempt from the flat-only rule because we do not author them.



5. Catalogue-first reuse + ecosystem submodule map (overview-level)

5.1 Catalogue-first discipline (§11.4.74)

Before scaffolding ANY new module/helper/utility, the contributor **MUST** survey the vasic-digital + HelixDevelopment orgs (the canonical inventory is `submodules-catalogue.md`) and **reuse** when an existing submodule covers ≥80% of the need, **extend upstream** when 80%+ matches but features are missing, and record Catalogue-Check: `reuse|extend|no-match <org/>repo>@<sha>` on the relevant tracker row [05_OV §6, §11.4.74]. Concrete applications already made for HelixVPN: containerization → extend `vasic-digital/containers` (§11.4.76); doc-sync → reuse `vasic-digital/docs_chain` (§11.4.106); QA → reuse `HelixDevelopment/helix_qa` (§11.4.27); design system → extend `OpenDesign` via `vasic-digital/helix_design` (§11.4.162).

5.2 Ecosystem submodule role map (the seam — depth in the sibling doc)

The 16 source docs predate the submodules/ incorporation, so the spec **MUST** wire each in or mark it not-applicable with a reason (§11.4.6, no silent drop) [05_OV §6.1]. This is the overview-level binding; the full integration depth (boot/health APIs, on-demand-infra invariant, docs_chain context schema, vision/QA harness wiring) lives in `helix-ecosystem-integration.md`.

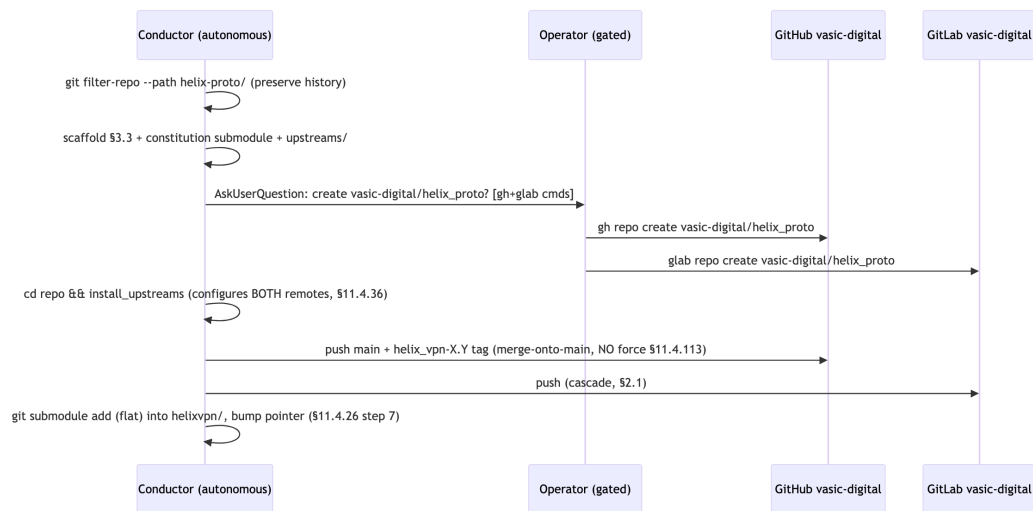
Submodule	HelixVPN role	Binding §	Phase
helix_design	Mandatory OpenDesign design system for all 3 apps × 8 platforms; decoupled reusable submodule (Volume 10)	§11.4.162	1+
containers	Sole container- orchestration layer: helixvpncctl deploy boot/health/compose primitives + on- demand integration- test infra	§11.4.76/.161	0–1
challenges	Challenge engine under the anti-bluff acceptance Challenges (MVP DoD as executable Challenges)	§11.4.27/.5/.69	1
helix_qa	Anti-bluff QA orchestrator: drives the 8 MVP-DoD criteria with captured evidence	§11.4.27/.107/.158	1
docs_chain	Mechanical sync of this spec set (final/ *.md → HTML/PDF/ DOCX) + workable- items SQLite DB  docs	§11.4.106/.65/.93	0+
security	Control-plane defensive libs (log PII redaction, security headers, at-rest secrets, SSRF-deny, edge privesc scan)	§11.4 (V5)	1
vision_engine	Video/screenshot evidence analysis for Flutter UI Challenges	§11.4.107/.158/.159	1–2
panoptic	UI automation + screenshot/video capture harness feeding vision_engine	§11.4.107/.154	1–2

Submodule	HelixVPN role	Binding §	Phase
doc_processor	Feature-map extraction → the per-feature Status ledger coverage cross-check	§11.4.153	1
llm_provider	Provider abstraction iff Phase-2 ships LLM-assisted policy authoring; NOT in the packet/control path	§11.4.74	2 (opt)
llm_orchestrator	Headless CLI-agent orchestration for the <i>development</i> loop; not a product runtime dependency	§11.4.70	dev-only
llms_verifier	Verifies any dev-loop LLM ("do you see my code?"); NOT a HelixVPN runtime component	§11.4.78	dev-only

Honest classification (§11.4.6, no silent drop): `llm_orchestrator` / `llms_verifier` are **dev-loop only** — HelixVPN ships no LLM in its data or control path; they support the autonomous build/QA loop and are correctly present as submodules (the project uses them to *build* itself) but are not linked into `helixd`/`helix-edge`. `llm_provider` is **conditional** (binds only if Phase-2 ships LLM-assisted policy authoring — Decision D-LLM-POLICY). Marking these runtime-N/A is the honest classification, never an omission.

6. Repo creation procedure (operator-gated, §11.4.66/.101)

Decision D-REPO-CREATE (operator-gated). Remote repo creation (`gh repo create` / `glab repo create on vasic-digital`) is high-blast-radius and **cannot** be done autonomously (§11.4.101 block-only rule). The extraction PR is prepared autonomously (filtered history, scaffold, submodule pointer staged); the actual repo-create + first push is surfaced to the operator with the exact `gh`/`glab` commands [05_OV §3.3]. **Recommended:** create all six extract-later repos in one batch at the Phase-1 exit gate so the release tags share the `helix_vpn-` prefix across every repo in one release (§11.4.151).



Each push is a fast-forward merge-onto-latest-main (§11.4.113 — **no force-push anywhere**), fanned out to every upstream (§2.1). After the push lands, `install_upstreams` is verified by `git remote -v | grep -c push` reporting the expected remote count (§11.4.36).

7. Release prefixing across every repo (§11.4.151)

Every release tag/version on the monorepo **and** every extracted reusable repo **MUST** carry the `<PREFIX>-` prefix, resolved deterministically as (1) `HELIX_RELEASE_PREFIX` from `.env` (authoritative; documented in the tracked `.env.example`), else (2) the lowercased snake_case root dir name = `helix_vpn` [05_OV §9, §11.4.151]. One release tags all repos with the **same** prefix in one batch (§6 recommendation) so a single greppable query enumerates the whole release surface across GitHub + GitLab:

```
git tag -l 'helix_vpn-*' # enumerates the whole
                        release across every repo
```

OCI images are tagged `<PREFIX>-<version>` to match. Version codes increment monotonically within the prefix, never reset, never skipped (§11.4.151).

8. Mapping to workable items (§11.4.93)

This document's deliverables become rows in the git-tracked SQLite SSoT (`docs/.workable_items.db`, §11.4.95). Representative seed items [05_OV §10]:

ATM-id (illustrative)	Item	Type	Phase	Dep
HVPN-050	Stand up helixvpn/ monorepo workspaces (Cargo / go.work / melos / buf)	Task	0	—
HVPN-056	Extract 6 reusable repos to vasic-digital (operator- gated, §6)	Task	1-exit	all code repos
HVPN-057	helix_design decoupled submodule scaffold + consumption recipe (Volume 10)	Task	1	HVPN-050
HVPN-058	helix-deps.yaml manifests + CM-OWNED- SUBMODULE-LAYOUT flat- graph audit	Task	1	HVPN-056
HVPN-059	Decoupling-invariant gate (CM-OWNED-SUBMODULE- DECOUPLING) per extracted repo	Task	1	HVPN-056

Each carries status/type/id on all surfaces (§11.4.148) and a comprehensive description (§11.4.148-D2).

9. Anti-bluff evidence plan (§11.4.5/.69/.107)

Claim	Captured-evidence proof (Phase)
Extraction preserves history (§1.2)	<code>git log --follow</code> on a sample file in the extracted repo shows pre-extraction commits; transcript under docs/qa/<run- id>/
No nested own-org submodule chains (§4)	CM-OWNED-SUBMODULE-LAYOUT audit of the extracted repos GREEN; mutation adding a nested .gitmodules own-org entry → FAIL
Decoupling holds (§3.2)	CM-OWNED-SUBMODULE-DECOUPLING GREEN; mutation injecting a HelixVPN hostname into helix_core → FAIL
Flat dependency-from- root (§4)	each consumer reaches its dep via root path; no submodule has an own-org .gitmodules entry

Claim	Captured-evidence proof (Phase)
install_upstreams configured both mirrors (§6)	git remote -v grep -c push == expected count post-clone; captured
Release prefix uniform (§7)	git tag -l 'helix_vpn-*' across every repo returns the same prefix; mutation creating an unprefixed tag → CM-RELEASE- PREFIX-NAMING FAIL

10. Decision callouts (options + recommendation — §11.4.66/.101)

id	Decision	Options	Recommendation
D-REPO-CREATE	When to extract reusable repos	(a) eager per-pillar; (b) one batch at Phase-1 exit	(b) — shared release prefix, stable seams (§1.2)
D-DESIGN-EXTRACT	When helix_design decouples	(a) extract-later like the six; (b) decoupled day-one	(b) — Volume 10 mandate (§11.4.162); independent of code churn
D-LLM-POLICY	Ship LLM-assisted policy authoring?	(a) yes (binds llm_provider); (b) no (stays dev-only)	Defer to Phase 2; default (b) until product-validated (§5.2)

These compose with the spine's decision register (D1–D8); none is resolved here without the operator where it is irreversible/high-blast-radius (§11.4.101).

Sources verified

- 05-repo-layout-tooling-and-helix-ecosystem.md §1 (monorepo-now/extract-later), §2 (file-by-file tree + workspace manifests), §3 (six decoupled repos + scaffold + deps manifest + creation procedure), §6 (ecosystem submodule role map + N/A justifications), §9 (release prefix), §10 (workable items), §12 (decision callouts) — [05_0V].
- 04_VPN_CLD/HelixVPN-Architecture-Refined.md §11 (repo layout), §4.1 (modular monolith / package-boundary discipline) — [04_ARCH].

- 04_VPN_CLD/HelixVPN-Phase1-MVP.md §2 (repo skeleton), HelixVPN-Phase0-Spike.md (surviving Transport trait + FFI extraction seams), HelixVPN-helix-ui-Flutter.md (runHelixApp flavors, Melos) — [04_P1] [04_P0] [04_UI].
- v09-research/_SYNTHESIS.md §3 (decisions/D5 helix-transport reuse), §6 (repo layout / reuse pillars), §8 (submodule wiring), §9 (constitution bindings) — [SYNTHESIS].
- MASTER_INDEX.md Volume 10 (helix_design decoupled submodule, §11.4.162) — [MASTER_INDEX].
- Constitution anchors: §11.4.28/.29/.31/.36/.74 (decoupled snake_case flat submodules + upstreams
 - deps manifest), §11.4.35 (visible reconciliation / canonical-root), §11.4.66/.101 (decision discipline / operator-gated repo create), §11.4.76/.161 (containers submodule sole orchestration
 - rootless), §11.4.93/.95 (workable items in git-tracked SQLite SSoT), §11.4.113 (no force-push), §11.4.151 (release prefix), §11.4.156 (no active CI), §11.4.162 (OpenDesign decoupled submodule), §11.4.6 (no-guessing — UNVERIFIED marks + N/A justifications), §11.4.5/.69/.107 (captured-evidence proofs in §9).